

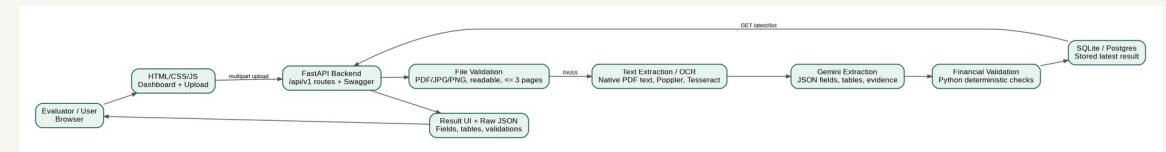
Document Intelligence

AI-powered extraction, validation and REST API platform

Problem and objective

Turn financial documents into structured, validated and retrievable JSON.

- Accept invoices and financial statements in PDF, JPG and PNG format.
- Validate upload constraints before any extraction work starts.
- Use OCR plus Groq to extract fields, tables and evidence, with Gemini as automatic fallback.
- Run deterministic financial checks and persist the result for dashboard/API retrieval.



Supported scope

Document type is selected manually; the app does not auto-classify documents.

- Invoice: header fields, parties, currency, totals and line items.
- Balance Sheet: header, periods, currency, assets, liabilities and equity.
- Profit & Loss: revenue/income, expenses, tax, profit and comparative periods.
- Cash Flow: operating, investing, financing cash flows, opening and closing cash.

Backend design

Small modules with clear responsibilities instead of one large file.

- FastAPI exposes /process, /documents, /documents/{name}, /health and Swagger docs.
- Validation service rejects unsupported, empty, corrupted or over-limit uploads.
- OCR service uses native PDF text first, then Poppler + Tesseract when OCR is required.
- Repository layer stores latest processed JSON in SQLite by default.

Extraction approach

Groq is the primary extractor; Gemini is the automatic fallback. Financial truth stays deterministic in Python.

- OCR/native text is sent with page markers so values can be grounded to page evidence.
- Prompt requires all meaningful visible fields and tables, not only a minimal schema.
- Missing or unreadable values remain null to avoid hallucination.
- Temporary 429/5xx/timeouts are retried; provider failure automatically falls back Groq → Gemini.

Financial validation

Checks are calculated in Python using only extracted values.

- Invoice: quantity x unit price, line-item sum, subtotal/tax/discount/total, cash change.
- Balance Sheet: assets reconcile with liabilities plus equity, per period when present.
- P&L: income, expenditure, net profit and appropriation checks when inputs exist.
- Cash Flow: operating + investing + financing + adjustments, then opening to closing cash.
- If required operands are missing, the check returns NOT_APPLICABLE.

Frontend and usability

Plain HTML/CSS/JS dashboard connected to the same FastAPI backend.

- Upload control with document-type selector and process action.
- Dashboard lists stored processed documents from the database-backed API.
- Result page shows file validation, extracted fields, line items, tables and checks.
- Missing values, failed validations and raw JSON are visible for evaluator review.

Testing, deployment and limitations

Prototype is deployable, explainable and designed for a 3-day assessment.

- Tests cover validation, OCR behavior, extraction schema, financial checks and API flow.
- Dockerfile installs Poppler and Tesseract for hosted OCR support.
- Environment variables hold secrets; .env is excluded from Git.
- Known limitations: OCR quality, table layout ambiguity, provider rate limits and no authentication.
- Production upgrades: managed Postgres, file storage, job queue, monitoring and stronger confidence method.